Mathematical Typesetting

Technical Writing Practical

1 Basics

Einstein's mass–energy equation is $E = mc^2$.

$$v = u + at$$

2 Fractions and Roots

$$x_i = \frac{\sqrt{a^2 + b^2}}{c_i}$$

3 Symbols and Math Fonts

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad \forall x \in \mathbb{R}\ \exists \epsilon > 0 \Rightarrow |x| < \epsilon$$

$$F(\theta) = \sum_{k=0}^{n} \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

4 Calculus

$$\frac{du}{dx} + \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \qquad \int_0^1 (3x^2 + 2x + 1)\, dx$$

5 Multi-line Equations

$$x = 5 \qquad x + y = 3 \tag{1}$$

$$x = 5 \qquad x + y = 3 \tag{2}$$

$$\begin{aligned}(x+y+z)^3 &= (x+y+z)(x+y+z)(x+y+z) \\ &= x^3 + y^3 + z^3 + 3x^2y + 3xy^2 + 3y^2z + 3yz^2 \\ &\qquad + 3x^2z + 3xz^2 + 6xyz\end{aligned}$$

6 Matrices

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \quad \begin{Vmatrix} 1 & 2 \\ 3 & 4 \end{Vmatrix}$$